# Simulation Guide

for *Investigating Lesion Registration Uncertainty in Dynamic Thermography Using Fiducial Leverage and Residuals* (IEEE ISBI 2027, under review) · **Wei-Che Hung**, Department of Systems and Control Engineering, University of Malta · 2 November 2026 · release v1.0.2-isbi2027. What is simulated, which script produces each figure and table, and how to run them; numbering follows the paper.

## 1. What the simulation tests

Four fiducial markers are registered frame by frame with an affine model, eq. (1). The paper predicts the registration error at a lesion point of interest (POI) from two quantities that every recording with four or more markers provides:

- the **leverage** $h_p$ of the POI, eq. (4), computed once from the marker and POI coordinates in frame-0;
- the **marker residuals** $R^{(k)}$ and their covariance $\hat{\Sigma}^{(k)}$, eq. (5), pooled over the sequence into $\Sigma_{\mathrm{eff}}$, eq. (6).

The prediction $\mathrm{RMS}_p = \sqrt{h_p \,\mathrm{tr}(\Sigma_{\mathrm{eff}})}$, eq. (7), extends to the intra-lesion error $\mathrm{RMS}_L$, eq. (8), and to the boundary area disagreement $\mathbb{E}[D_A]$, eq. (9); each is compared with the Monte Carlo value from the same trials.

## 2. Protocol (paper Sec. 3.4)

| Item | Value |
|---|---|
| Image | 640 × 480 pixels |
| Markers | four; *baseline* configuration, and *expanded* configuration with each marker moved 1.5 times farther from the marker centroid |
| POIs | 65 (a 9 × 7 grid plus two interior points); ray of 100 POIs from the marker centroid beyond the marker convex hull, σ = 1 pixel |
| Motion per trial | affine: rotation ±5°, translation ±10 pixels, scaling ±2 % |
| Marker noise | isotropic Gaussian, σ ∈ {0.5, 1, 2, 4} pixels, added to the moved markers |
| Trials | 10 000 per noise level, seed 0 (bootstrap band of Fig. 4: seed 1, 1000 resamples) |
| Lesion polygons | triangle, circle, rectangle of equal area at the marker centroid (δ = 0); circles at normalised offsets δ ∈ {0.5, 1.0, 1.5}; σ = 1 pixel, repeated at σ = 2 for Sec. 4.2 |

Every script uses the same trial procedure and draw order: draw the motion, move markers and POIs, add marker noise, fit the affine map back to frame-0, record POI errors and marker residuals. The prediction uses residuals and leverage only; the truth uses the known POI positions.

## 3. Scripts, outputs and paper objects

Commands run from the repository root. Script names keep the draft numbering (`fig1` → Fig. 2, `fig2` → Fig. 3, `fig3` → Fig. 4). Only `fig1_leverage_map.py` asserts pinned numbers before drawing; the others write their statistics to `results/`.

| Script | Produces | Paper object |
|---|---|---|
| `src/analysis1/fig1_leverage_map.py` | leverage map, worked frame (seed 1, σ = 1): $h = 1.7806$ at $p_1 = (40, 40)$ | Fig. 2 |
| `src/analysis1/fig2_probe_rms.py` | predicted vs simulated RMS for the 65 POIs, both configurations, four noise levels; `results/fig2_probe_rms_stats.json` | Fig. 3, Table 1a |
| `src/analysis2/fig3_polygon_rms.py` | ray sweep with the 95 % simultaneous band; six lesion polygons with 95 % confidence intervals; `results/fig3_polygon_rms_stats.json` | Fig. 4, Table 1a |
| `src/analysis3/area_error.py` | predicted and simulated area disagreement $D_A$ for the six polygons; `results/table1_area_error_stats.json` | Table 1b |
| `src/analysis3/table1_render.py` | the 18 numbers of Table 1 from the three stats files | Table 1 |

Modules: `src/common/affine.py` (geometry, motion, noise), `src/analysis1/mc_core.py` (trial procedure, Fig. 3). Tests: `python -m unittest discover -s <dir>` for `src/common`, `src/analysis1`, `src/analysis2`, `src/analysis3`.

**Run.**

```
python src/analysis1/fig1_leverage_map.py
python src/analysis1/fig2_probe_rms.py --seed 0 --trials 10000
python src/analysis2/fig3_polygon_rms.py --seed 0 --trials 10000 --sigma 1
python src/analysis2/fig3_polygon_rms.py --seed 0 --trials 10000 --sigma 2 \
    --out results/fig3_polygon_rms_sigma2.pdf --stats results/fig3_polygon_rms_sigma2_stats.json
python src/analysis3/area_error.py --seed 0 --trials 10000 --sigma 1
python src/analysis3/table1_render.py
```

Requirements: Python 3.10 or later, NumPy, Matplotlib (SciPy for the analysis3 tests only). Figures go to `paper/figures/`, statistics to `results/`; Table 1 is read from those files at render time.

**Code availability and contact.** This repository holds the guide; the scripts are released here after the review round. Wei-Che Hung, wei-che.hung@um.edu.mt or wolfmib@gmail.com.